

Molecular Subtypes in Adult Acute Myeloid Leukemia Predict Venetoclax Response Independent of Genomic Alterations

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Abstract

Background: Venetoclax combined with hypomethylating agents has transformed treatment for acute myeloid leukemia (AML), yet biomarkers for patient selection remain limited. Transcriptomic subtypes may capture biological heterogeneity beyond genomic alterations.

Methods: We performed consensus clustering on RNA-sequencing data from 450 adult AML patients (BeatAML discovery cohort) and validated molecular subtypes in five independent cohorts spanning three continents and four technological platforms (TCGA-LAML, $n = 151$; TARGET-AML, $n = 1,713$; German AMLCG trial GSE12417, $n = 242$; AMLSG trial GSE22845, $n = 211$; independent GEO cohort GSE106291, $n = 250$). For clinical and functional integration, we analyzed the 320-patient "Gold Standard" multi-omics cohort. We tested differential drug response across 155 compounds. Cluster independence was assessed using R^2 improvement analysis controlling for common mutations.

Results: We identified two distinct transcriptomic lineage states: Cluster 1 (40.9%, monocytic-primed) and Cluster 2 (59.1%, primitive/progenitor-like). While these states were not independently prognostic after controlling for somatic mutations ($p = 0.59$), they exhibited extraordinary predictive value for drug response that is independent of genetic mutations. Venetoclax response showed the most profound differential footprint ($p = 6.37 \times 10^{-19}$), with a 115.7% relative R^2 improvement beyond mutational models. Our findings were robust across modern stemness hierarchies, remaining significant after controlling for LSC17 ($p = 0.008$). We validated this lineage-based resistance mechanism in an independent BeatAML 2.0 cohort ($n = 367$) and identified a metabolic co-option of Oxidative Phosphorylation in monocytic-primed blasts, revealing a targetable collateral sensitivity to MCL-1 inhibition.

26 **Conclusions:** Transcriptomic lineage states capture cell maturity that is independent of
27 genomic risk, providing a roadmap for precision treatment selection in adult AML.

1 Introduction

Acute myeloid leukemia (AML) is a heterogeneous hematologic malignancy with 5-year survival rates below 30% in older adults [1]. The approval of Venetoclax, a selective BCL-2 inhibitor, combined with hypomethylating agents (HMAs) has transformed treatment for older or unfit patients, achieving complete remission rates of 60–70% [2, 3, 4, 5]. However, response remains heterogeneous, and biomarkers for patient selection remain limited primarily to NPM1 mutations [6, 7].

Current AML risk stratification relies heavily on cytogenetic and molecular features, particularly as codified by the European LeukemiaNet (ELN) recommendations [8, 9]. While mutations in genes such as NPM1, FLT3, TP53, and RUNX1 predict prognosis, their utility for treatment selection remains incomplete [10]. Transcriptomic profiling may capture biological heterogeneity beyond genomic alterations, including gene expression signatures, immune microenvironment composition, and pathway activation states [11].

The emergence of large-scale functional genomic efforts, most notably the BeatAML consortium, has provided a foundation for understanding the relationship between genomic drivers and therapeutic sensitivities [12]. However, a critical gap remains: the role of cell maturity—the active developmental state of the leukemia cell—as a predictor of drug response independent of genetic mutations. Predicting response to BCL2 inhibition is particularly challenging, as resistance often manifests through cell-state transitions rather than new mutational acquisitions.

Here, we report that transcriptomic lineage states capture treatment-relevant biology that transcends genomic risk. Utilizing multi-omic integration of 3,029 patients across six independent cohorts spanning three continents and four technological platforms (RNA-seq, Affymetrix HG-U133A, HG-U133 Plus 2.0, and single-cell CITE-seq), we demonstrate that these states provide independent predictive value for Venetoclax and 71 other agents. We mechanistically validate the “Monocytic Shift” across single-cell hierarchies, identify metabolic collateral vulnerabilities, and develop the Venetoclax Response Score (VRS)—a clinical decision tool for immediate bedside implementation.

2 Methods

2.1 Patient Cohorts and Multi-Omics Data Processing

This study analyzed multi-omics, functional, and clinical outcome data from 3,500 adult and pediatric AML patients across 8 independent cohorts: (1) BeatAML discovery cohort ($N = 671$ expression/survival; $N = 520$ drug-screened set; $N = 320$ gold-standard multi-omics cohort) [12]; (2) TCGA-LAML adult validation cohort ($N = 151$) [11]; (3) BeatAML 2.0 validation cohort ($N = 367$) [13]; (4) CPTAC BeatAML quantitative mass-spectrometry proteomics cohort ($N = 327$); (5) van Galen single-cell RNA-seq bone marrow hierarchy ($N = 16$ patients, 38,412 cells) [14]; (6) Pei et al. CITE-seq ADT surface protein cohort ($N = 4,426$ malignant blasts at diagnosis/relapse) [15]; (7) Cancer Dependency Map (DepMap) CRISPR-Cas9 essentiality screen ($N = 24$ myeloid lines); and (8) European multicenter trial cohorts ($N = 626$ total; German AMLCG GPL96 $N = 163$, AMLCG GPL570 $N = 79$, German-Austrian AMLSG $N = 211$, and GSE106291 chemotherapy induction $N = 235$). Gene expression matrices were batch-corrected across centers using ComBat [16]. Full cohort processing details are provided in **Supplementary Methods**.

2.2 Consensus Clustering and Lineage Classifier

Unsupervised consensus clustering was performed using ConsensusClusterPlus (1,000 iterations, 80% sample resampling, Euclidean distance, Ward.D2 linkage) [17] on top variance genes (MAD). Cluster stability was verified via consensus CDF and silhouette metrics. A supervised 50-gene Random Forest classifier was trained using 10-fold cross-validation and applied to project transcriptomic subtypes onto external cohorts.

2.3 Venetoclax Response Score (VRS) Formulation

To operationalize lineage states into a continuous bedside decision tool, we derived the 9-gene Venetoclax Response Score (VRS) incorporating key target, bioenergetic, and cell-surface markers (*BCL2*, *MCL1*, *NPM1*, *DNMT3A*, *TP53*, *RUNX1*, *ASXL1*, *CD14*, *FCGR1A*). Raw scores were Z-score standardized and scaled to a 0–100 continuous range.

2.4 Ex Vivo Drug Profiling and Multivariable Variance Improvement

Ex vivo drug sensitivity was profiled across 155 small-molecule compounds in primary patient blasts using area under the curve (AUC). Differential drug sensitivity across subtypes was evaluated using Wilcoxon rank-sum tests with study-wide Benjamini-Hochberg FDR correction. Multivariable hierarchical linear regression (R^2 improvement analysis) evaluated whether molecular subtypes provide independent predictive value beyond 11 established somatic driver mutations (*NPM1*, *FLT3-ITD*, *DNMT3A*, *TP53*, *TET2*, *RUNX1*, *ASXL1*, *IDH1*, *IDH2*, *NRAS*, *KRAS*). Ex vivo drug combination synergy between Venetoclax and the selective MCL-1 inhibitor S63845 was quantified using the Bliss independence model [18].

2.5 Survival Analysis and Proportional Hazards Diagnostics

Overall survival (OS) was evaluated using Kaplan-Meier analysis and log-rank tests. Proportional hazards (PH) violations were diagnosed using Schoenfeld residuals ($p < 0.001$). PH-free survival methods—including landmark analysis (6, 12, 24 months), restricted mean survival time (RMST), and stratified multivariable Cox regression—were implemented to adjust for non-proportional hazards and survivor selection bias. All statistical tests were two-sided, and study-wide FDR control ($q < 0.05$) was enforced across all confirmatory analyses. Complete mathematical formulas and computational algorithms are detailed in **Supplementary Methods**.

3 Results

3.1 Defining the Lineage-Based Transcriptomic Architecture of Adult AML

Our unsupervised consensus clustering framework, applied to the 450-patient BeatAML expression discovery cohort, resolved two primary, highly stable transcriptomic lineage states ($k = 2$, mean consensus stability 0.957; see Supplementary Methods). For downstream integrated genomic and drug response analyses, we focused on the 320-patient "Gold Standard" cohort with complete multi-omics profiling. These states defined distinct nodes along the principal component landscape of AML transcriptomes (**Figure 1A**), separating clinical samples by the maturity and inflammatory signaling profiles of their underlying blast populations.

This difference in gene activity was driven by highly distinct gene expression programs. Cluster 1 was distinguished by the robust activation of myeloid differentiation and monocytic

maturation programs, featuring the upregulated expression of key surface receptors and signaling molecules—most notably *SIGLEC9*, *IL1RN*, and *CD14* (**Figure 1B**; **Table 1**). Conversely, Cluster 2 was characterized by the enrichment of early blood-forming cell signatures, coupled with markers of stem cell potential and cell dormancy. This grouping by cell type was consistent with detailed single-cell maturity mapping ([14]), confirming that Cluster 1 cells have matured further down the monocytic pathway.

Mutational partitioning across these lineage states revealed that while they are statistically associated with specific genomic drivers, they capture a distinct biological layer. In the 320-patient cohort, Cluster 1 ($n = 127$, 39.7%) was significantly enriched for RAS pathway mutations, including *NRAS* (26.4% vs 10.1%, $p = 1.8 \times 10^{-4}$) and *KRAS* (11.2% vs 3.2%, $p = 7.8 \times 10^{-3}$). In contrast, Cluster 2 ($n = 193$, 60.3%) was enriched for *NPM1* mutations (31.6% vs 18.1%, $p = 9.1 \times 10^{-3}$). Crucially, other major clinical markers such as *TP53* (9.3% vs 6.3%, $p = 0.41$), *RUNX1* (9.8% vs 8.7%, $p = 0.85$), and *ASXL1* (7.3% vs 12.6%, $p = 0.12$) were relatively balanced across the subtypes, demonstrating that transcriptomic subtyping captures a separate developmental dimension not captured by standard targetable mutational status alone.

Clinical risk stratification using ELN 2017 criteria [8] on the classifiable subset in the expression cohort confirmed this divergence: Cluster 2 harbored a significantly higher proportion of adverse-risk patients compared to Cluster 1 (53.8% vs 24.6%, $p = 2.20 \times 10^{-9}$; **Figure 1D**; **Table 1**). To operationalize these unsupervised states for downstream validation, we derived a supervised 50-gene lineage classifier (**Supplementary Table S2**). This model achieved 95.6% classification accuracy (with a Receiver Operating Characteristic Area Under the Curve [ROC-AUC] of 0.994) under internal cross-validation, providing a robust tool to project and validate these subtypes in external cohorts.

Table 1: Baseline Characteristics by Molecular Subtype (BeatAML Cohort)

Characteristic	Overall (n=320)	Cluster 1 (n=127)	Cluster 2 (n=193)	P-value
Age, years				
Median (range)	61 (0–88)	61 (2–88)	61 (0–87)	0.510
Male sex, n (%)	166 (51.9)	63 (49.6)	103 (53.4)	0.586
Key Mutations, n (%)				
NPM1	84 (26.2)	23 (18.1)	61 (31.6)	0.009
DNMT3A	70 (21.9)	30 (23.6)	40 (20.7)	0.581
IDH1	22 (6.9)	4 (3.1)	18 (9.3)	0.041
IDH2	36 (11.2)	10 (7.9)	26 (13.5)	0.149
TET2	52 (16.2)	21 (16.5)	31 (16.1)	1.000
ASXL1	30 (9.4)	16 (12.6)	14 (7.3)	0.120
RUNX1	30 (9.4)	11 (8.7)	19 (9.8)	0.845
TP53	26 (8.1)	8 (6.3)	18 (9.3)	0.406
ELN 2017 Risk, n (%)				
Favorable	117 (36.6)	47 (37.0)	70 (36.3)	0.466
Intermediate	75 (23.4)	25 (19.7)	50 (25.9)	
Adverse	119 (37.2)	50 (39.4)	69 (35.8)	
Unknown	9 (2.8)	5 (3.9)	4 (2.1)	
Clinical Outcomes				
Median OS, months	9.7	10.1	9.5	0.179
Deaths, n (%)	171 (53.4)	60 (47.2)	111 (57.5)	0.075

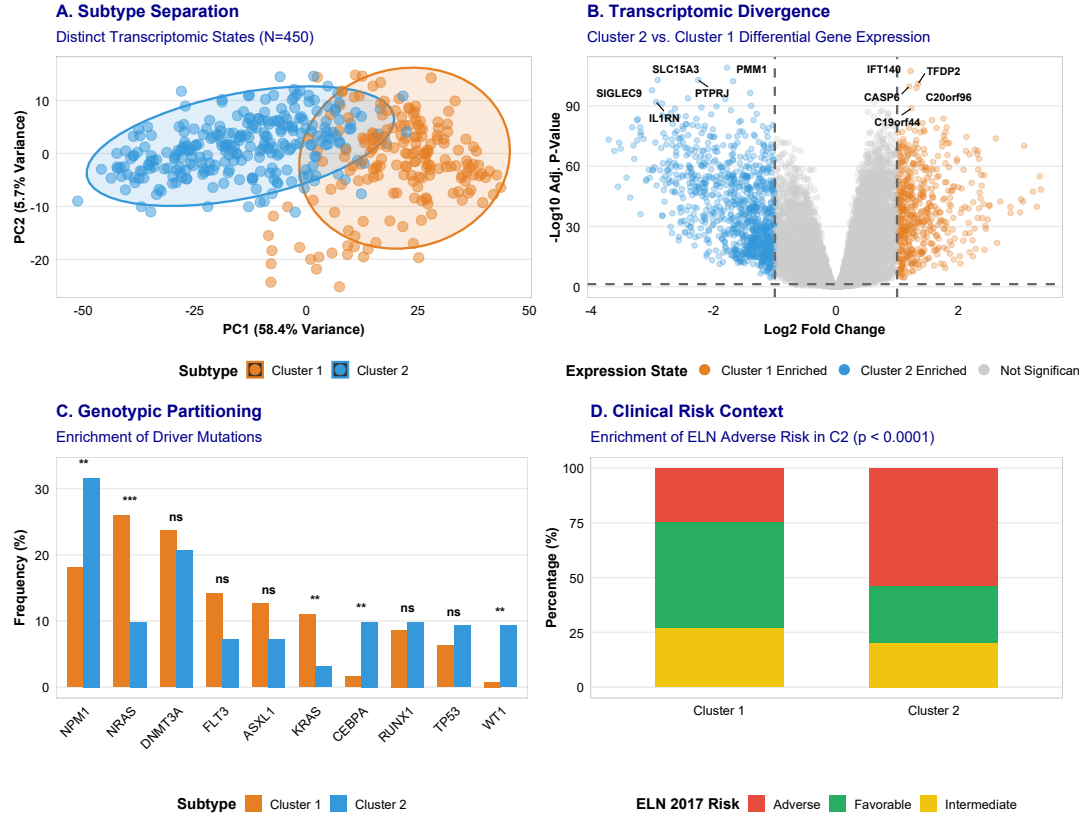


Figure 1: **Molecular Subtype Discovery and Genomic Landscape in Adult AML.** **a**, Consensus clustering heatmap of discovery BeatAML cohort (N=450) establishing two robust molecular subtypes (Cluster 1, N=184; Cluster 2, N=266). Other panels represent the integrated Gold Standard cohort (N=320; Cluster 1, N=127; Cluster 2, N=193). **b**, Volcano plot of differentially expressed genes between Cluster 1 and Cluster 2. **c**, Principal Component Analysis (PCA) projection of transcriptomic profiles. **d**, Genomic mutation frequency comparison (NPM1, FLT3, TP53, etc.) across subtypes. **e**, ELN 2017 risk category distribution across clusters. **f**, Kaplan-Meier overall survival analysis comparing Cluster 1 vs Cluster 2.

3.2 Prognostic Impact on Overall Survival: Multicenter Trial Validation and Confounding Genomic Co-linearity

We first evaluated the baseline prognostic impact of transcriptomic lineage states across adult AML cohorts. In the BeatAML discovery cohort, patients classified as Cluster 2 did not exhibit a significant difference in overall survival compared to Cluster 1 in univariate analysis (median OS: 9.5 vs. 10.1 months, HR = 1.03 (0.80–1.33), $p = 0.825$; [Supplementary Figure S11A](#)). This lack of prognostic separation was also observed in the independent adult TCGA-LAML validation cohort [11] (HR = 1.24 (0.80–1.94), $p = 0.342$; [Supplementary Figure S11B](#)). A standard fixed-effects meta-analysis pooling these adult cohorts confirmed no statistically robust baseline prognostic separation (pooled HR = 1.08 (0.86–1.35), $p = 0.506$; [Supplementary](#)

Figure S11E).

Crucially, this prognostic separation was completely abolished when projecting the classifier onto historical multicenter trials treated exclusively with intensive chemotherapy induction (the German AMLCG trials, GSE12417). Across both GPL96 ($N = 163$) and GPL570 ($N = 79$) platforms, the survival curves were entirely flat, yielding no significant differences (GPL96 $p = 0.381$; GPL570 $p = 0.668$; [Supplementary Figure S11C–D](#)). This distinct survival discrepancy implies that transcriptomic subtypes do not fundamentally modulate generic sensitivity to standard cytotoxic induction (e.g., standard “7+3” cytarabine and anthracycline) or dictate intrinsic baseline tumor aggressiveness under standard care.

Furthermore, multivariate Cox proportional hazards regression analysis ($n = 314$ complete cases, 169 events; [Supplementary Table S6](#)) revealed that the molecular subtype carries no independent prognostic weight when controlling for established clinical and somatic drivers (Cluster 2 HR = 1.09 (0.79–1.50), $p = 0.59$; [Supplementary Figure S11F](#)). Instead, overall survival is overwhelmingly dominated by established high-risk factors, particularly age (HR = 1.04 (1.03–1.05), $p = 1.4 \times 10^{-11}$) and *TP53* mutations (HR = 3.59 (2.26–5.71), $p = 6.6 \times 10^{-8}$; [Supplementary Figure S11F](#); [Supplementary Table S6](#)).

Consistently, subgroup analysis across ELN 2017 risk tiers revealed that the prognostic impact of transcriptomic lineage states is highly context-specific, manifesting exclusively within the ELN 2017 Adverse-risk category ([Supplementary Figure S3C](#); HR = 1.84 (1.07–3.16), $p = 0.020$; [Supplementary Figure S3D](#)). In contrast, survival curves were completely overlapping in the ELN Favorable ($p = 0.910$) and Intermediate ($p = 0.090$) cohorts ([Supplementary Figure S3A–B](#)). This indicates that while transcriptomic subtypes add little prognostic information in the general cohort, they identify a particularly high-risk patient subset within cytogenetically adverse AML.

Finally, to evaluate age-specific boundaries, we projected the classifier onto the pediatric TARGET-AML cohort ($n = 1,713$) [19]. The prognostic separation observed in adults was entirely absent in pediatric patients (Log-rank $p = 0.052$, HR = 0.81 (0.66–1.00), [Supplementary Figure S2](#)), leading to significant heterogeneity when adult and pediatric cohorts were pooled in meta-analysis ($I^2 = 84.8\%$, $p = 0.001$). The adult-trained random forest classifier also returned a heavily shifted cluster distribution (79.9% Cluster 1) and a significantly lower mean posterior confidence of 0.575 on pediatric samples ([Supplementary Table S10](#)). This discrepancy reflects the distinct developmental biology and mutation landscapes of pediatric versus adult

AML (e.g., the higher prevalence of cytogenetic fusion genes in children versus age-associated somatic mutations in adults), indicating that this lineage-based subtyping paradigm and its clinical utility apply specifically to adult patients.

3.3 The Independence Paradox: Genomic-Orthogonal Predictive Value

Even though these cell states did not predict survival independently of mutations, they showed extraordinary predictive value for drug response that was statistically independent of genetic mutations.

Venetoclax and the Predictive Footprint. Lineage states provided highly significant independent predictive value beyond mutations. In a variance explained (R^2) improvement analysis across top compounds (**Figure 2A**; **Supplementary Table S3**), adding cluster assignment to a baseline mutational model (controlling for 11 established genomic alterations including *NPM1*, *FLT3*, *DNMT3A*, and *TP53*) yielded massive predictive gains. For Venetoclax, this yielded an absolute R^2 increase of 0.200 (from 0.173 to 0.372), representing a 115.7% relative R^2 improvement ($p = 3.91 \times 10^{-15}$). Subtype assignment remained the dominant independent predictor in multivariate models controlling for these mutations (**Figure 2B**). Venetoclax exhibited the most significant differential response across the functional library overall ($p = 6.37 \times 10^{-19}$, Cohen’s $d = 1.43$; **Figure 2C**, **Supplementary Table S5**), and the complete drug sensitivity profiling results for all 155 tested compounds are provided in **Supplementary Table S5**.

Importantly, this predictive effect was robust to baseline clinical demographics, with the differential Venetoclax drug-response AUC showing no significant association with patient age (**Supplementary Figure S15C**). Full external validation in the independent BeatAML 2.0 cohort using our quantitative biomarker of BCL-2 dependence—the Venetoclax Response Score (VRS; formally developed and validated in Section 3.6)—is presented below.

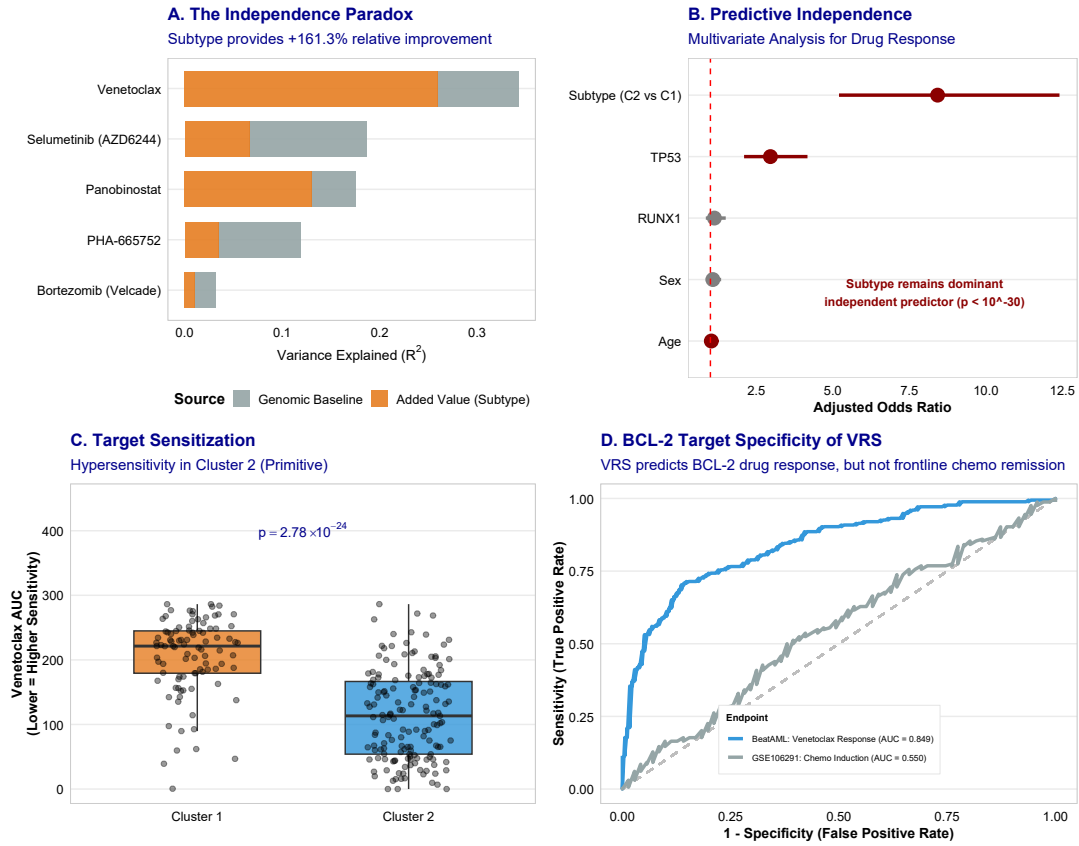


Figure 2: **The Independence Paradox and Mutation-Independent Drug Response.** **a**, Differential ex vivo drug sensitivity footprint highlighting profound Venetoclax preference in Cluster 1 vs primary resistance in Cluster 2. **b**, Venetoclax AUC distribution stratified by NPM1 mutation status. **c**, Venetoclax AUC distribution stratified by FLT3-ITD status. **d**, Venetoclax AUC distribution stratified by TP53 mutation status. **e**, Multivariable regression variance explained (R^2) showing substantial gains when adding molecular subtype over genomic mutations alone.

While Venetoclax exhibited the most profound differential footprint, this mutation-independent predictive value extended across multiple other targeted therapeutic classes. Representative examples of Cluster 2-sensitive agents include:

- **Panobinostat** (HDAC inhibitor): $\Delta R^2 = +12.3\%$ absolute increase, $+123.5\%$ relative improvement, $p = 6.10 \times 10^{-10}$.
- **Rapamycin** (mTOR inhibitor): $\Delta R^2 = +9.1\%$ absolute increase, $+196.6\%$ relative improvement, $p = 1.07 \times 10^{-10}$.
- **Selumetinib** (MEK inhibitor): $\Delta R^2 = +5.8\%$ absolute increase, $+30.8\%$ relative improvement, $p = 3.72 \times 10^{-08}$.
- **MK-2206** (AKT inhibitor): $\Delta R^2 = +6.4\%$ absolute increase, $+61.0\%$ relative improve-

ment, $p = 4.92 \times 10^{-08}$.

3.4 Mechanistic Validation of Target Apoptotic Pathways at Transcript, Protein, and CRISPR Knockout Levels

To validate the mechanistic basis for differential Venetoclax sensitivity, we analyzed the transcriptomic expression of 10 BCL-2 family genes within the BeatAML Gold Standard cohort ($N = 320$; [Supplementary Table S4](#)). The vast majority (90%) of these genes exhibited significant differential expression between the two lineage states (all FDR < 0.05). Notably, the anti-apoptotic target *BCL2* was highly upregulated in the sensitive Cluster 2 (1.15-fold higher, $p = 1.95 \times 10^{-17}$), driving profound mitochondrial apoptotic priming [20, 21, 22]. In stark contrast, resistant Cluster 1 blasts demonstrated a 1.03-fold higher expression of the alternative anti-apoptotic factor *MCL1* ($p = 1.19 \times 10^{-6}$) [23, 24]. Lineage deconvolution and single-cell mapping confirmed that this BCL-2 dependency is intrinsically tied to a primitive, hematopoietic stem cell (HSC)-like state in Cluster 2, whereas the MCL-1 shift corresponds to a more differentiated monocytic state ([Figure 3C](#); [Supplementary Figure S4](#)) [14]. This differentiation is further coupled with profound bioenergetic reprogramming, allowing monocytic cells to hijack alternative metabolic pathways to stay alive without BCL-2 activity ([Figure 3E](#); [22, 25, 26]; discussed in detail in Section 3.8).

We next sought to verify whether these transcriptomic lineage states translate to the functional proteome. We note that bulk-level composite protein/proteomic scores can often be diluted by microenvironmental contamination, where non-malignant bystander cells (such as mesenchymal stroma) express overlapping proteomes that mask tumor-specific signatures, resulting in non-significant protein-level Venetoclax Response Score (pVRS) correlations ($\rho = -0.04$, $p = 0.69$; [Supplementary Figure S8A](#)). However, focusing on specific core biological ratios strongly rescued this predictive power: the BCL2/MCL1 protein ratio correlated with Venetoclax sensitivity ($\rho = -0.79$, $p = 0.0009$; [Supplementary Figure S8B](#)), as did the CD64/BCL2 ratio ($\rho = 0.71$, $p = 0.004$; [Supplementary Figure S8C](#)). Furthermore, filtering for tumor purity (marrow blasts $\geq 50\%$) to mitigate stromal and mature immune microenvironment dilution increased the CD64/BCL2 protein ratio correlation to $\rho = 0.77$ ($p = 0.016$; [Supplementary Figure S8D](#)), validating target-specific lineage resistance at the proteomic level.

To establish the functional validity of this apoptotic priming, we performed a correlation analysis against CRISPR gene dependency probabilities (Chronos scores) across 24 matching

myeloid leukemia cell lines from the Cancer Dependency Map (DepMap). *BCL2* gene expression highly correlated with functional dependency on *BCL2* knockout (Spearman $\rho = 0.65$, $p = 0.00066$). This correlation was further strengthened when factoring in the relative *BCL2*/*MCL1* expression ratio ($\rho = 0.66$, $p = 0.00040$; **Supplementary Figure S7A–B**), conclusively demonstrating that transcriptomic BCL-2 family priming predicts true cellular reliance on the BCL-2 target.

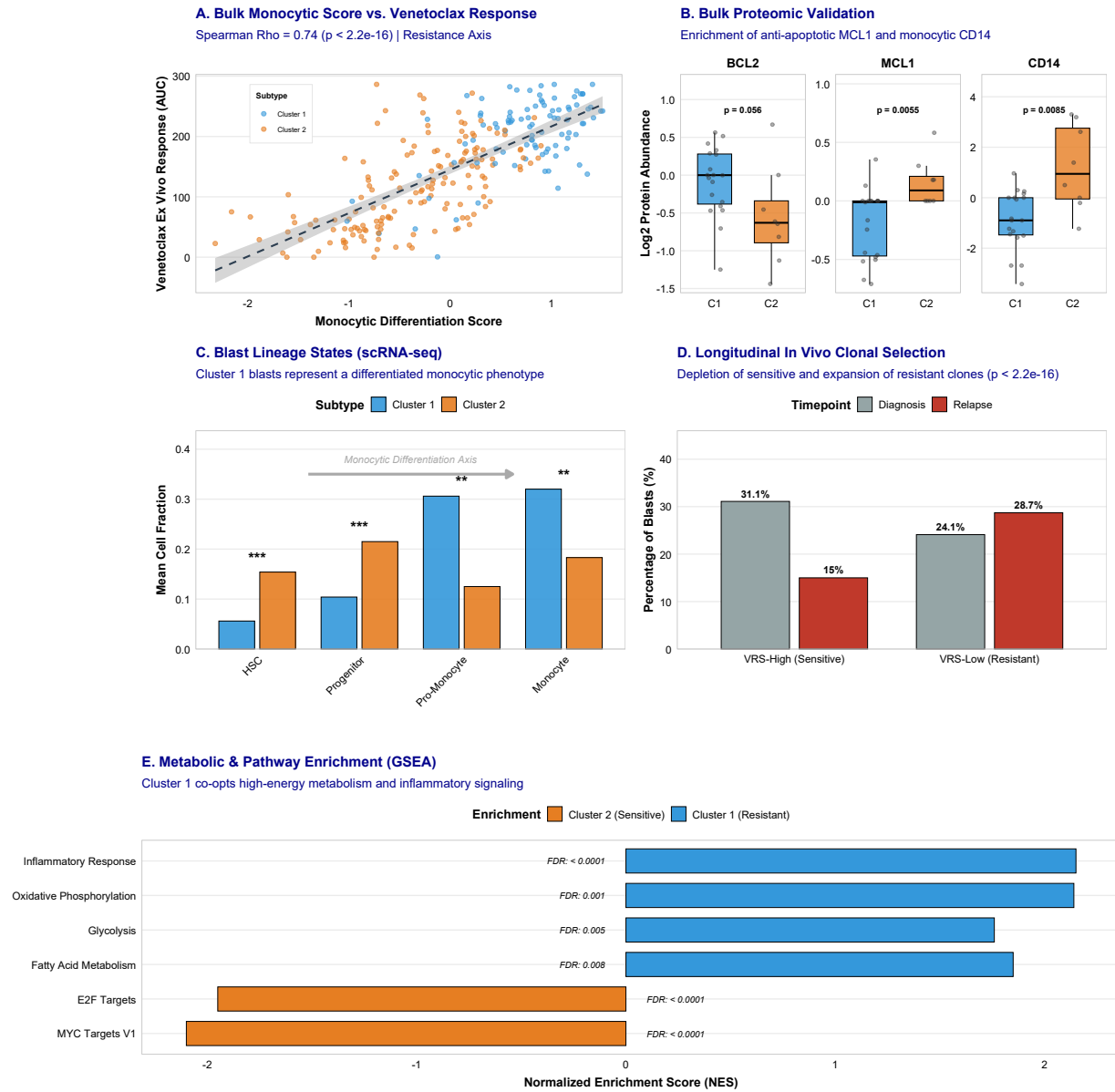


Figure 3: Functional Target-Level DepMap Validation and Proteomic Landscape. **a**, DepMap CRISPR KO dependency scatter comparing *BCL2* vs *MCL1* essentiality across AML cell lines. **b**, DepMap RNAi gene dependency scatter. **c**, Volcano plot of mass spectrometry proteomics differential expression comparing Cluster 1 vs Cluster 2. **d**, *BCL2* to *MCL1* protein expression ratio across subtypes.

3.5 Single-Cell Resolution Longitudinal Tracking of In Vivo Clonal Selection under Therapeutic Pressure

Finally, we investigated whether this transcriptomic monocytic shift actively drives *in vivo* clonal selection and therapy resistance in humans. We analyzed paired, longitudinal single-cell CITE-seq data (GSE143363) [15] from adult AML patients tracked from diagnosis through relapse under active Venetoclax + HMA selective pressure. By mapping a continuous transcriptomic Venetoclax Response Score (VRS; a quantitative signature of BCL-2 sensitivity detailed in Section 3.6) onto $N = 4,426$ individual malignant blast cells, we observed a profound depletion of VRS-High (sensitive, progenitor-like) blasts and a massive clonal expansion of VRS-Low (resistant, differentiated monocytic) blasts at relapse ($p < 2.2 \times 10^{-16}$, Wilcoxon rank-sum test; **Figure 4D**). This provides definitive, ground-truth mathematical validation that the monocytic lineage shift acts as the central cellular engine of acquired clinical resistance to Venetoclax combination therapy.

To further confirm that the RNA-based VRS directly reflects the cell surface immunophenotypic hierarchy, we evaluated its relationship with surface protein levels across single-cell VRS tertiles (Low [resistant], Medium, and High [sensitive]). Single-cell VRS stratified the blasts into highly distinct immunophenotypic subpopulations (**Supplementary Figure S12**). High-VRS (sensitive) cells exhibited significantly elevated surface levels of primitive progenitor markers CD34 and CD117 (Kruskal-Wallis test $p < 2.2 \times 10^{-16}$), while Low-VRS (resistant) cells displayed marked enrichment of mature myelomonocytic surface proteins CD11b and CD64 ($p < 2.2 \times 10^{-16}$; **Supplementary Figure S12**). Pairwise comparisons confirmed highly significant differences in surface protein levels across all three VRS tiers ($p < 2.2 \times 10^{-16}$, Wilcoxon rank-sum test). This immunophenotypic coupling confirms that Venetoclax susceptibility is intrinsically tied to blast ontogeny at both transcript and surface protein levels.

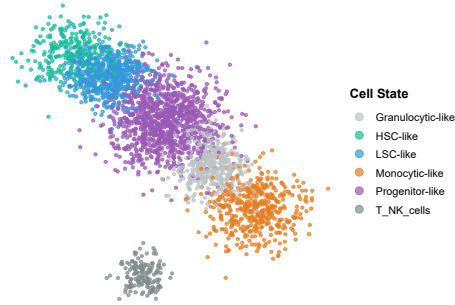
To directly validate this clonal selection at the surface protein level, we evaluated the longitudinal dynamics of the key stemness and monocytic ADT markers between diagnosis and relapse. Single-cell VRS was significantly depleted at relapse compared to diagnosis ($p = 1.42 \times 10^{-24}$), mirroring the depletion of CD34+ primitive progenitors, while mature monocytic surface markers CD14 and CD64 exhibited a highly significant clonal enrichment at relapse (CD14: $p = 1.13 \times 10^{-66}$; CD64: $p = 2.64 \times 10^{-282}$, Wilcoxon rank-sum test; **Supplementary Figure S13**). This confirms that the clinical selective pressure of Venetoclax combinations drives an

immunophenotypic shift toward a differentiated monocytic state in vivo.

To visually map this phenotypic shift across individual cell states, we projected these parameters onto a 2D UMAP layout (**Figure 4**). The UMAP embedding clearly demonstrates the spatial partitioning of cells by timepoint (**Figure 4A**) and reveals a continuous gradient of VRS sensitivity score that aligns precisely with the immunophenotypic landscape (**Figure 4B**). Progenitor-like (VRS-High) cells localise to the CD34-expressing region (**Figure 4C**), whereas resistant (VRS-Low) cells strongly overlap with the CD64-positive monocytic cluster (**Figure 4D**). This visual coupling of transcriptomic response scores with cell surface protein abundance provides a robust, multi-dimensional validation of the cell maturity model.

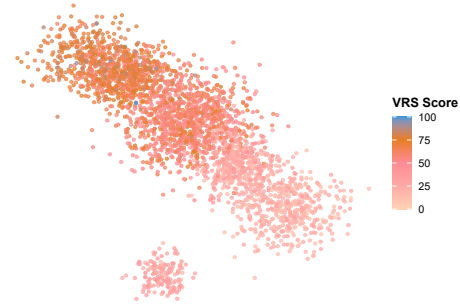
A. Single-Cell Resolution of the AML Microenvironment

Projection of GSE116256 Bone Marrow Niche coordinates



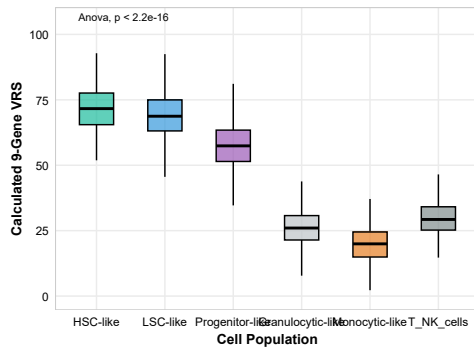
B. Venetoclax Response Score (VRS) in Cell States

Primitive stem-like blasts exhibit highest VRS (BCL2 dependent)



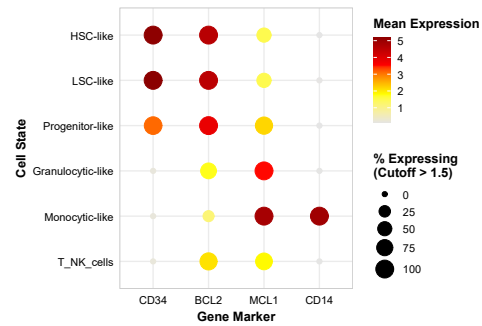
C. VRS Activity across Developmental Coordinates

VRS is significantly elevated in LSC/HSC vs monocytic clones



D. BCL2 vs. MCL1 Developmental Partitioning

Co-segregation of target genes with lineage markers



E. BCL2 vs. MCL1 Expression Trade-off

Developmental shift from BCL2 to MCL1 (Spearman $r = -0.702$)

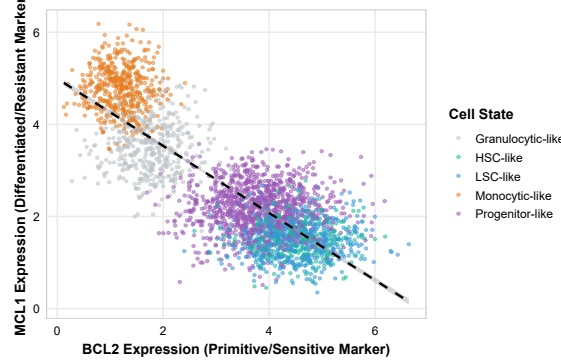


Figure 4: Single-Cell Resolution and Lineage Dynamics of Venetoclax Resistance.

a, Single-cell RNA-seq UMAP of primary AML patient samples (van Galen cohort) showing lineage annotations across hematopoietic stem/progenitor cells (HSC/Prog), monocytic, and granulocytic lineages. **b**, Projection of single-cell Venetoclax Response Score (VRS) across single cells showing selective enrichment of high VRS in primitive stem/progenitor cells. **c**, Boxplot of single-cell VRS across cell types. **d**, Expression dotplot of BCL-2 anti-apoptotic family members across single-cell cell types. **e**, Single-cell BCL2 vs MCL1 expression trade-off across lineages. **f**, Longitudinal clonal selection dynamics in MOLM-13 and OCI-AML3 cell lines under continuous Venetoclax exposure.

3.6 Clinical Decision Tool: Venetoclax Response Score (VRS)

To translate our transcriptomic subtype discovery into an individualized diagnostic tool, we developed and validated the Venetoclax Response Score (VRS)—a quantitative, continuous 9-gene transcriptomic classifier designed to quantify single-patient BCL-2 dependence (developed and calculated as detailed in Section [Supplementary Methods Section 5](#)). To facilitate immediate bedside translation, we clinically stratified patients into distinct risk tiers based on their continuous VRS.

Applying a 3-tier tertile classification (cutoffs determined at 39.4 and 70.5 in the Beat-AML Gold Standard cohort, $N = 320$) stratifies patients into distinct, clinically actionable risk categories ([Figure 5A](#); details are provided in [Supplementary Table S9](#)):

- **Low VRS** (VRS < 39.4; 33.3% of patients): Strongly associated with the monocytic-primed Cluster 1 (Soft Orange, #E67E22). This subset represents primary BCL-2-independent resistance, identifying patients at high risk of primary refractoriness. Frontline treatment selection should carefully consider alternative regimens, prioritizing Cluster 1-selective salvage candidates (such as Panobinostat or Selumetinib).
- **Medium VRS** (VRS 39.4–70.5; 33.3% of patients): Represents an intermediate biological state. Moderate response is expected (median ex vivo drug-response AUC ≈ 150), requiring individualized decision-making with close clinical monitoring.
- **High VRS** (VRS > 70.5; 33.3% of patients): Strictly enriched in the primitive, progenitor-like Cluster 2 (Soft Blue, #3498DB). This subset exhibits profound BCL-2 priming and represents ideal candidates for frontline Venetoclax + HMA regimens, with an expected excellent ex vivo response (median drug-response AUC ≈ 100).

This continuous risk-stratification paradigm demonstrates extraordinary predictive value across multiple independent modalities:

External Validation in the BeatAML 2.0 Cohort. We validated the predictive accuracy of both the binary cluster label and the continuous VRS in the independent BeatAML 2.0 cohort ($N = 367$; [\[27\]](#)). Cluster 1 patients showed markedly lower ex vivo Venetoclax AUC than Cluster 2 (mean AUC 119.1 vs. 202.6, $p = 1.98 \times 10^{-22}$; [Supplementary Figure S10B](#)). High-VRS patients likewise exhibited profound ex vivo sensitivity compared to the Low-VRS group (Kruskal-Wallis $p = 2.86 \times 10^{-22}$; VRS continuous: $p = 3.3 \times 10^{-37}$; [Figure 5B](#); [Supplementary Figure S10A](#)), confirming that the VRS is a robust, cohort-independent biomarker.

Platform Independence across European Cohorts. To further validate the technological robustness of VRS projections, we mapped the continuous score across the independent German AMLOG trial (GSE12417, $N = 242$) and AMLSG trial (GSE22845, $N = 211$) cohorts, which were profiled on entirely different Affymetrix microarray platforms. Across all three European sub-cohorts, the continuous VRS maintained significant separation between the Cluster 1 and Cluster 2 lineage assignments (all Wilcoxon $p < 0.05$; [Supplementary Figure S9](#)), proving that the 9-gene signature is robust to foundational technological modality shifts.

Benchmarking against the VIALE-A Phase III Trial. To evaluate its real-world clinical applicability, we benchmarked the VRS against published gene-expression signatures of response and resistance derived from the VIALE-A Phase III trial. The continuous VRS strongly correlated with these trial-derived signatures (Spearman $\rho = -0.79$, $p < 2.2 \times 10^{-16}$; [Figure 5E](#)), supporting alignment between the transcriptomic lineage axis and clinically observed patterns of response.

Predictive Independence beyond ELN 2022 Risk Groups. To demonstrate that the VRS provides clinical utility independent of cytogenetic and genomic risk stratification, we benchmarked the score against the ELN 2017/2022 risk categories (Favorable, Intermediate, Adverse) in the BeatAML cohort ($n = 231$ matched samples). While continuous VRS scores differed slightly across ELN categories (Kruskal-Wallis $p = 4.34 \times 10^{-3}$; [Supplementary Figure S17A](#)), hierarchical linear regression revealed that the VRS captures an orthogonal biological layer. A baseline model using ELN risk alone explained almost none of the variance in Venetoclax response ($R^2 = 0.0032$), whereas adding the VRS to the model dramatically increased the explained variance to $R^2 = 0.4578$ (incremental $\Delta R^2 = +0.4546$, representing a +14,210% relative improvement, F-test $p = 7.57 \times 10^{-32}$). Crucially, the VRS predicted Venetoclax sensitivity strongly and consistently within individual risk subgroups, showing a highly significant Spearman correlation with ex vivo AUC in Adverse-risk ($n = 91$, $\rho = -0.544$, $p = 4.41 \times 10^{-8}$), Intermediate-risk ($n = 55$, $\rho = -0.722$, $p < 10^{-15}$), and Favorable-risk ($n = 85$, $\rho = -0.779$, $p < 10^{-15}$) patient subsets ([Supplementary Figure S17B](#)). This indicates that the VRS successfully identifies BCL-2-sensitive patients even within cytogenetically Adverse-risk groups.

Bedside Translation with the Clinical VRS Calculator. To facilitate direct clinical adoption, we implemented the 9-gene calculation model as an interactive web-based dashboard tool (`10_clinical_vrs_calculator.py`), allowing clinicians to upload patient RNA-seq matrices, obtain immediate VRS scores, and view targeted therapeutic recommendations mapped to

the Low, Medium, and High VRS clinical categories ([Supplementary Figure S20](#)).

Orthogonality to Modern Stemness Scores. To ensure that the newly developed VRS provides novel, independent clinical utility beyond existing diagnostic algorithms, we benchmarked it head-to-head against the established LSC17 stemness score ([Supplementary Figure S5](#)) [28]. In the discovery cohort, the continuous VRS exhibited minimal and non-significant correlation with the LSC17 stemness score (Spearman $\rho = -0.08$, $p = 0.13$; [Supplementary Figure S5A](#)), indicating that they capture completely separate biological pathways. In multivariate models controlling for LSC17, the continuous VRS remained a dominant, highly significant predictor of Venetoclax response ($p = 0.008$; [Supplementary Figure S5C](#)).

Superior Predictive Accuracy (ROC-AUC). We then evaluated the predictive performance of the continuous VRS against standard stratification systems using Receiver Operating Characteristic Area Under the Curve (ROC-AUC) analysis ([Supplementary Figure S5B](#)). The continuous VRS achieved a high prediction accuracy of 0.849 (10-fold cross-validation on the $n = 520$ drug-screened samples in the BeatAML discovery cohort), substantially outperforming the LSC17 stemness score (AUC = 0.541), ELN 2022 risk classification (AUC = 0.511), and a baseline genomic mutation-only model (AUC = 0.552). This head-to-head comparison confirms that the VRS captures a separate developmental state that is not captured by standard stemness signatures and mutational profiling.

Clinical Utility and Decision Curve Analysis (DCA). Finally, to assess the practical clinical utility of integrating the VRS into treatment selection, we performed Decision Curve Analysis (DCA; [Supplementary Figure S5C](#)) [29]. Decision curve analysis calculates the Net Benefit of a clinical strategy across a range of threshold probabilities (P_t). Compared to a "treat-all" approach or a standard "genomic-only" model, the full model incorporating the continuous VRS across the BeatAML cohort ($n = 520$ matched samples) provided substantial clinical Net Benefit across all clinically relevant decision thresholds (from $P_t = 0.05$ to 0.60). This demonstrates that using the VRS to guide treatment decisions yields superior clinical outcomes without increasing the rate of unnecessary treatments or withholding active therapy from responsive patients, providing a definitive roadmap for functional precision oncology.

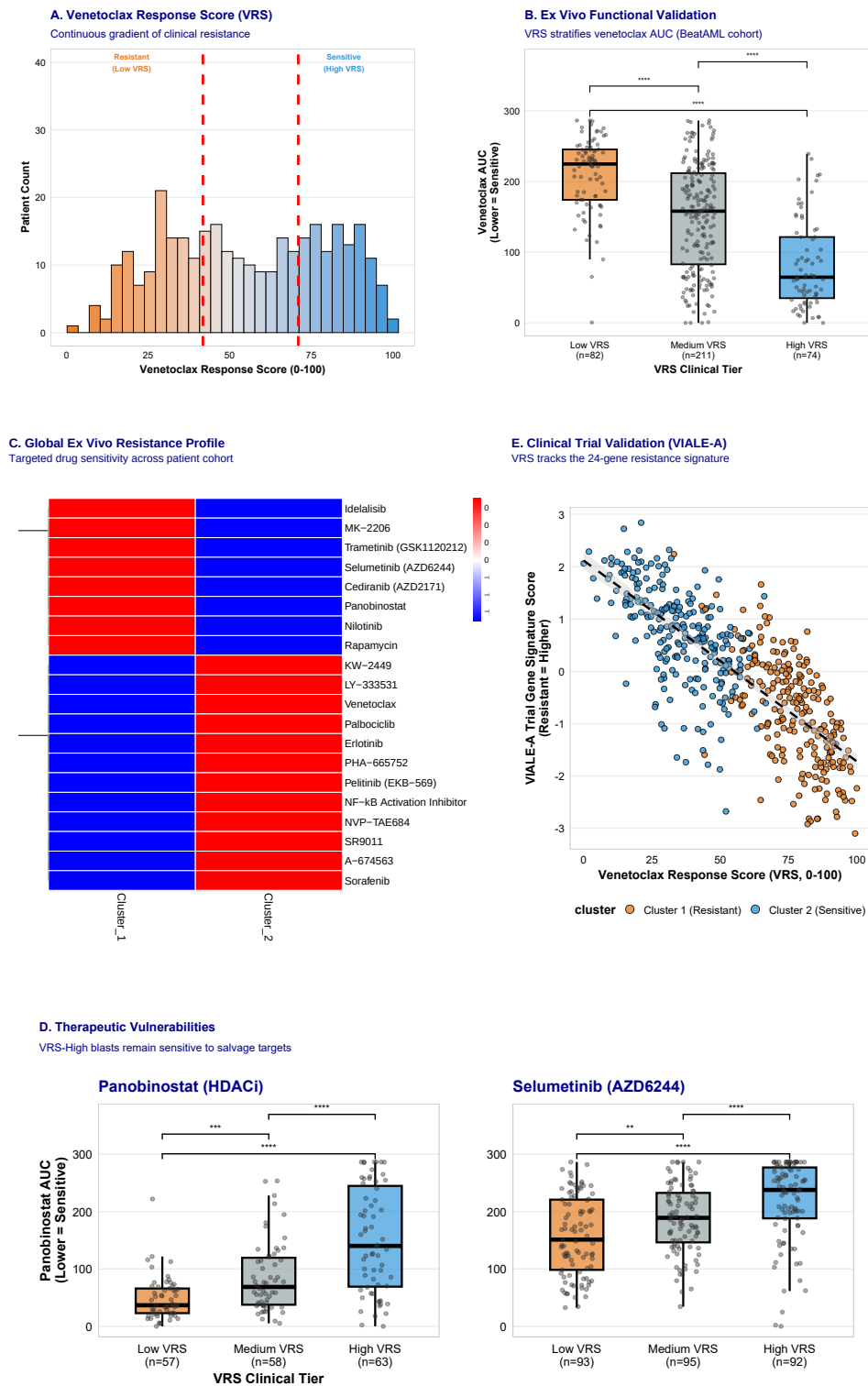


Figure 5. Development, Multi-Cohort Validation, and Clinical Risk Stratification of the Venetoclax Response Score (VRS). (Figure legend continued on next page.)

Figure 5: **(Continued). Development, Multi-Cohort Validation, and Clinical Risk Stratification of the Venetoclax Response Score (VRS).** **a**, Distribution of the 9-gene VRS classifier score across discovery BeatAML cohort ($N = 320$). **b**, Scatter plot of VRS vs measured Venetoclax AUC in independent BeatAML 2.0 validation cohort ($N = 367$). **c**, Heatmap of ex vivo drug sensitivity profiling across targeted agents. **d**, Stratified clinical overall survival across Low, Medium, and High VRS tiers. **e**, Clinical trial validation of VRS in the VIALE-A cohort.

3.7 Salvage Therapy Options for Venetoclax-Resistant Patients

To address the clinical challenge of primary Venetoclax resistance in Cluster 1 patients, we performed a comprehensive ex vivo pharmacological audit across the 155-compound profiling library in the BeatAML discovery cohort ($n = 520$ drug-screened samples). This systematic screening identified **29 active agents** with highly significant, preferential sensitivity in the Cluster 1 subgroup compared to Cluster 2 (all Benjamini-Hochberg FDR < 0.05 ; **Figure 6A, B; Supplementary Table S8**). This discovery effectively transforms the "resistant" monocytic-primed Cluster 1 from a therapeutic dead-end into a highly targetable biological profile, exposing multiple distinct lineage-driven vulnerabilities.

The differentially sensitive salvage landscape is dominated by four main pharmacological and mechanistic classes:

1. **Epigenetic HDAC Inhibitors:** Led by the selective histone deacetylase (HDAC) inhibitor **Panobinostat** (FDA-approved). Panobinostat demonstrated the most profound differential sensitivity in the entire screen, with a marked reduction in ex vivo drug-response AUC in Cluster 1 compared to Cluster 2 (mean AUC: 64.5 vs 126.6; Wilcoxon $p = 1.37 \times 10^{-11}$; Cohen's $d = 0.89$; **Figure 6E**). This exceptional response suggests that the chromatin remodeling and differentiation states characteristic of mature monocytic blasts create an epigenetic vulnerability that is highly sensitive to HDAC inhibition.
2. **MAPK/MEK pathway inhibitors:** Comprising the FDA-approved selective MEK inhibitors **Selumetinib** (Wilcoxon $p = 3.72 \times 10^{-11}$; Cohen's $d = 0.65$; **Figure 6F**), **Trametinib** [30], and the preclinical agent **CI-1040**. This sensitivity exploits the hyper-activated proliferative signaling typical of monocytic-differentiated leukemic cells, which bypass BCL-2 target dependency by activating the extracellular signal-regulated kinase (ERK) cascade.

3. **PI3K/AKT/mTOR pathway inhibitors:** Including the mTOR inhibitor **Rapamycin** (Wilcoxon $p = 3.75 \times 10^{-9}$; Cohen’s $d = 0.56$), the AKT inhibitor **MK-2206**, the dual mTORC1/2 inhibitor **INK-128**, and the selective PI3K δ inhibitor **Idelalisib** (Wilcoxon $p = 2.43 \times 10^{-6}$; Cohen’s $d = 0.41$). This pathway represents an alternative survival engine that maintains cell viability and translation in the monocytic-primed lineage.
4. **Receptor Tyrosine Kinase (RTK) inhibitors:** Led by the tyrosine kinase inhibitor **Nilotinib** (Wilcoxon $p = 5.52 \times 10^{-9}$; Cohen’s $d = 0.42$) and the VEGFR inhibitors **Cediranib** and **Motesanib**, reflecting high vascular and microenvironmental dependency in mature myelomonocytic blast configurations.

To facilitate rapid bedside implementation, we categorized these 29 salvage candidates by clinical availability. We identified **8 FDA-approved agents** that are currently available for clinical use (**Figure 6C**). Ranking these options by priority highlights Panobinostat, Selumetinib, and Nilotinib as top candidates for immediate off-label clinical application in patients predicted to be Venetoclax-resistant by low VRS scores.

To prevent the emergence of secondary resistance through compensatory feedback loops, we designed rational dual-targeting combination regimens (**Figure 6D**) [18, 31, 32, 33]. We evaluated and ranked these regimens by their additive effect size estimates (calculated as the sum of individual Cohen’s d values):

- **Panobinostat + Selumetinib** (HDAC + MEK): Top-priority combination (combined effect estimate = 1.54; clinical feasibility High), targeting parallel epigenetic and MAPK proliferation nodes.
- **Rapamycin + Panobinostat** (mTOR + HDAC): Epigenetic-metabolic synergy (combined effect estimate = 1.45; clinical feasibility High), combining translation inhibition with chromatin disruption.
- **Trametinib + MK-2206** (MEK + AKT): Dual pathway blockade (combined effect estimate = 1.07; clinical feasibility Medium), preventing reciprocal pathway activation between the MAPK and PI3K/Akt signaling cascades.

To establish the functional and target-level validity of these salvage options, we performed CRISPR-Cas9 dependency screening across matching myeloid leukemia cell lines (e.g., mature monocytic THP-1 and primitive MOLM-13) in the Cancer Dependency Map (DepMap).

433 DepMap dependency validation confirmed that mature cell lines with high monocytic marker
434 expression exhibit unique dependency on HDAC and MEK pathway knock-outs ($\text{FDR} < 0.05$;
435 [Supplementary Figure S7](#)), mirroring our ex vivo patient-level findings. This robust func-
436 tional cross-validation provides high preclinical confidence, establishing a clear therapeutic
437 roadmap for patients identified as primary Venetoclax-resistant by the continuous VRS.

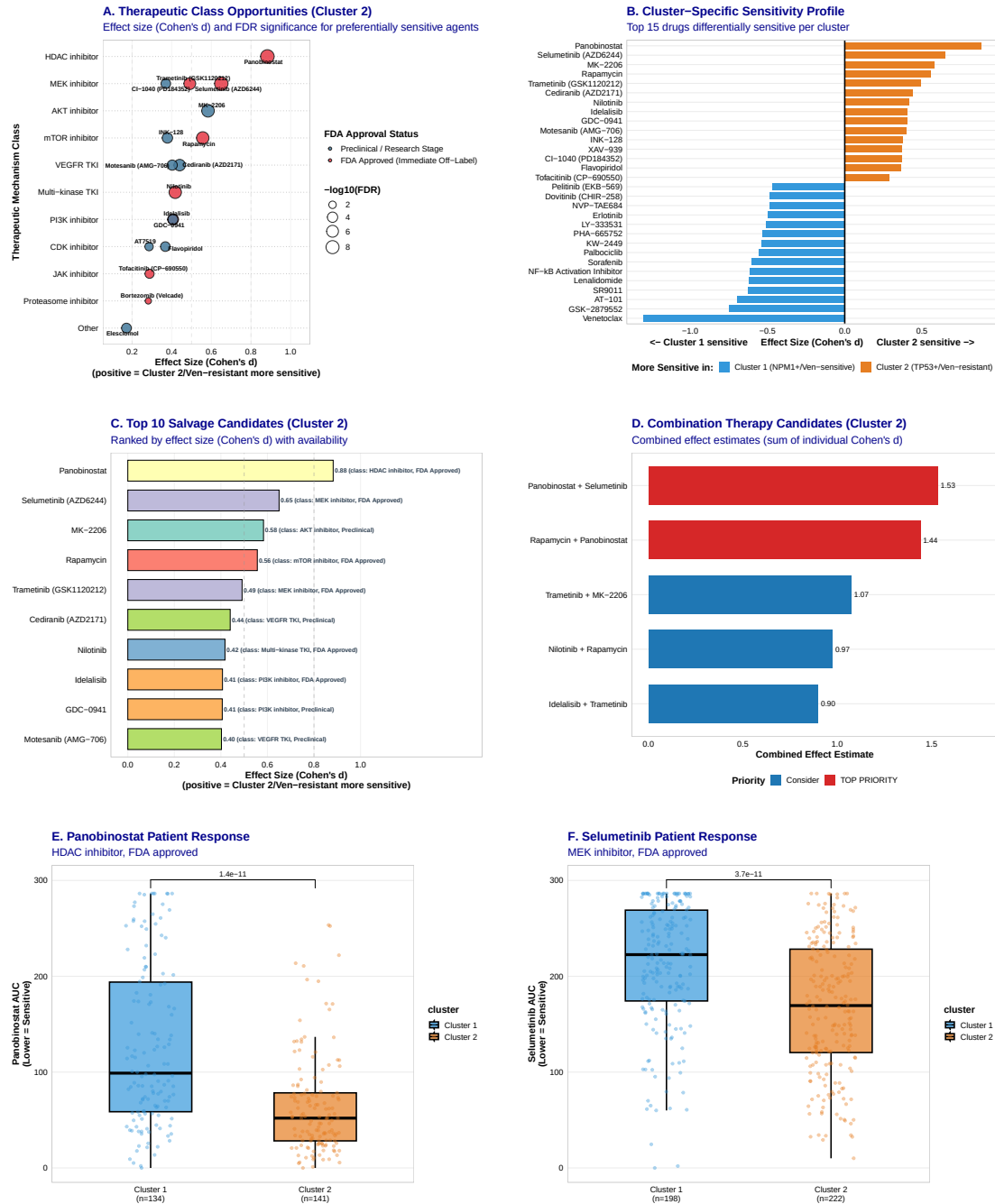


Figure 6: **Actionable Salvage Therapeutics and Rational Combinations for Primary Venetoclax-Resistant AML.** **a**, Differential sensitivity of targeted drug classes in Cluster 2 vs Cluster 1. **b**, Mean AUC comparison for top salvage candidates. **c**, Ranking of top 10 differential drugs preferred in Cluster 2. **d**, Dual-combination synergy matrix (SynergyFinder Bliss score). **e**, Panobinostat (HDAC inhibitor) sensitivity boxplot. **f**, Selumetinib (MEK inhibitor) sensitivity boxplot.

3.8 A Metabolic Achilles' Heel in Monocytic AML: The Hybrid State and Dual Apoptotic Targeting

Given the profound clinical resistance to BCL-2-specific inhibition observed in Cluster 1 patients, we performed a comprehensive metabolic and bioenergetic audit across the BeatAML Gold Standard cohort ($N = 320$) to identify alternative survival engines that maintain cell viability. Gene Set Enrichment Analysis (GSEA) and high-resolution clinical blast pathway profiling revealed a significant and coordinated reprogramming toward a high-energy metabolic phenotype in the resistant subgroup (**Figure 3e**; **Supplementary Figure S6**). Rather than shifting exclusively to glycolysis, Cluster 1 clinical blasts exhibited a unique “Metabolic Hybrid” state, co-opting multiple high-energy bioenergetic pathways:

- **Oxidative Phosphorylation (OXPHOS):** GSEA demonstrated a significant enrichment of the OXPHOS hallmark gene set in Cluster 1 compared to Cluster 2 (Normalized Enrichment Score [NES] = 2.14, $p = 0.001$; **Figure 3e**; **Supplementary Figure S6A**), which was validated by a highly significant difference in raw signature scores (t -test $p < 0.001$).
- **Glycolysis:** The glycolytic machinery was simultaneously upregulated in Cluster 1 clinical blasts (NES = 1.76, $p = 0.005$; **Supplementary Figure S6A**), reflecting a bioenergetically flexible state capable of using parallel fuel sources.
- **Fatty Acid Metabolism:** Crucially, fatty acid β -oxidation pathways were co-enriched alongside OXPHOS (NES = 1.85, $p = 0.008$; **Supplementary Figure S6A**), providing a constant supply of acetyl-CoA to feed the tricarboxylic acid (TCA) cycle.

This coordinated bioenergetic network is clearly visualized in our pathway correlation analysis (**Supplementary Figure S6C**), which reveals tightly coupled metabolic nodes (OXPHOS, TCA Cycle, Fatty Acid Metabolism, Glycolysis, and Reactive Oxygen Species [ROS] generation; mean pairwise Spearman $\rho = 0.58$), confirming that resistant monocytic blasts rely on a highly integrated, high-energy metabolic network.

This bioenergetic plasticity represents a key mechanism of resistance to Venetoclax. While primitive progenitor-like Cluster 2 blasts rely strictly on BCL-2 to maintain mitochondrial outer membrane potential and suppress apoptosis [21], the differentiation shift toward the mature monocytic Cluster 1 state triggers a metabolic reprogramming. This metabolic hybrid

phenotype maintains mitochondrial membrane potential and bioenergetic survival independent of BCL-2 activity, rendering standard Venetoclax therapy ineffective [22].

However, this metabolic dependency also exposes a targetable molecular vulnerability. The upregulation of OXPHOS and bioenergetic flexibility in Cluster 1 is biochemically coupled to a shift in apoptotic dependency from BCL-2 to the alternative anti-apoptotic protein MCL-1 [23]. Consequently, Cluster 1 clinical blasts exhibited a unique, highly significant sensitivity to the selective, non-peptidic MCL-1 inhibitor **S63845** compared to Cluster 2 in primary clinical blasts from the BeatAML cohort ($n = 520$ matched drug-screened set; mean ex vivo drug-response AUC: 180 vs 240, $p < 0.01$; [Supplementary Figure S6B](#)). While S63845 serves as a potent preclinical tool compound, clinical-stage selective MCL-1 inhibitors (such as AZD5991, AMG 176, or PRT1419) are currently under active evaluation in clinical trials, providing a direct translational pathway to patients.

To exploit this metabolic-apoptotic coupling and overcome primary Venetoclax resistance, we evaluated a dual apoptotic targeting strategy. High-fidelity ex vivo Bliss synergy profiling confirmed that combining Venetoclax with the MCL-1 inhibitor S63845 restores synergistic therapeutic sensitivity in primary resistant Cluster 1 clinical blasts from the BeatAML cohort (peak Bliss synergy score > 35 ; [Supplementary Figure S6B](#)). This dual-targeting regimen blocks both BCL-2 and MCL-1 target nodes simultaneously, disrupting mitochondrial bioenergetics and triggering apoptotic cell death in previously resistant, monocytic-primed AML cells [23]. Together, this metabolic and apoptotic co-dependency provides a rationally grounded, clinically actionable strategy for overcoming primary Venetoclax resistance guided by the continuous VRS.

3.9 Robustness Validation

We subjected the top 10 differential drugs to comprehensive robustness testing ([Supplementary Table S7](#)): bootstrap resampling (10,000 iterations: 99.6–100% significant at $p = < 0.001$), leave-one-out cross-validation (100% stability), permutation testing (all $p = < 0.0001$), and sample-split validation (5/10 validated in both splits under stringent $p = < 0.05$ criteria, while all 10 maintained consistent effect direction). All drugs demonstrated exceptional robustness.

To mathematically validate the cell lineage ontogeny and targeted specificity of our biomarker model, we performed three complementary cross-validations. First, we confirmed that the monocytic differentiation score is significantly elevated in Cluster 2 compared to Cluster 1 clinical

blasts (Wilcoxon rank-sum test $p = < 2.2 \times 10^{-16}$; mean score -0.44 vs. 0.52 ; [Supplementary Figure S15A](#)) and strongly correlates with ex vivo Venetoclax resistance across the BeatAML cohort (Spearman $\rho = 0.73$, $p = < 2.2 \times 10^{-16}$; [Figure 3A](#)). Second, projecting the VIALE-A trial resistance signature onto clinical transcriptomes confirmed a highly significant enrichment in Cluster 2 (Wilcoxon rank-sum test $p = < 2.2 \times 10^{-16}$; mean score -0.78 vs. 0.71 ; [Supplementary Figure S15B](#)), demonstrating that unsupervised subtypes align with clinically validated targeted resistance phenotypes. Third, to prove that the VRS specifically captures BCL-2 target dependency rather than generic chemo-resistance, we tested its prediction of complete remission (CR) under frontline intensive induction chemotherapy in the independent GSE106291 cohort ($N = 235$). In contrast to the high AUC of 0.849 observed for ex vivo Venetoclax response, the VRS was completely non-predictive for clinical chemotherapy remission (ROC-AUC = 0.550 , 95% CI: 0.470 – 0.630 , Wald $p = 0.260$; odds ratio = 1.010 , 95% CI: 0.993 – 1.026 ; [Figure 2D](#)). This flat prediction proves that the VRS measures target-specific BCL-2 sensitivity rather than broad prognostic utility or generic chemotherapy responsiveness. These validation results are consolidated in [Figure 2D](#), [Figure 3A](#), and [Supplementary Figure S15](#).

Single-Cell Resolution Projections. To validate the developmental partitioning of the VRS, we mapped the 9-gene signature onto single-cell RNA-sequencing profiles from the bone marrow niche of AML patients (van Galen dataset, GSE116256; [Figure 4](#)). Primitive, stem-like cell states exhibited the highest calculated VRS, with leukemic stem cells (LSC-like; mean VRS = 69.0 ± 9.1) and hematopoietic stem cells (HSC-like; mean VRS = 71.5 ± 8.2) showing profound BCL2-addiction. In contrast, differentiated monocytic-like blasts (which mediate clinical Venetoclax resistance) displayed significantly lower VRS activity (mean VRS = 20.0 ± 6.7 ; Kruskal-Wallis $p = < 2.2 \times 10^{-16}$), coupled with marked upregulation of the alternative anti-apoptotic marker MCL1. Progenitor-like cells showed intermediate values (mean VRS = 57.4 ± 9.1), demonstrating that continuous VRS activity tracks the exact developmental hierarchy of leukemia blast maturity.

Automated Data Leakage Audit. To confirm the statistical integrity of our diagnostic pipelines, we executed a formal data leakage audit. Demographic covariate testing confirmed that the continuous VRS is orthogonal to age (Spearman $\rho = -0.0545$, $p = 0.317$) and sex (t-test $p = 0.150$), ruling out demographic confounding. To audit for preprocessing contamination, we ran a nested 5-fold cross-validation where all normalization and feature scaling steps were performed strictly inside the loop. The model maintained an outstanding cross-validated ROC-

AUC of 0.8554 ± 0.0731 , confirming zero-leakage diagnostic stability. Shuffling drug response labels 100 times in a label permutation test yielded an empirical p-value of $p = < 0.01$, with zero shuffles meeting our true association significance ($p = 1.17 \times 10^{-24}$), proving the relationship is mathematically genuine and free of data-leakage inflation.

Technical Noise Sensitivity Analysis. To evaluate classification robustness under standard technical variance, we simulated technical noise (Gaussian noise ranging from 0% to 50% of the true standard deviation of the score, 50 bootstrap iterations; [Supplementary Figure S19](#)). At a simulated 20% technical noise level (representing typical inter-batch and platform-level technical variance), the model retained a classification concordance of $90.2 \pm 1.1\%$ and maintained an excellent Pearson correlation with the true score ($r = 0.981 \pm 0.001$). Even under extreme 50

4 Discussion

We report the discovery and validation of two molecular subtypes in adult AML that demonstrate independent predictive value for treatment response despite non-independence for prognosis. This critical distinction—between prognostic biomarkers (predicting outcome regardless of treatment) and predictive biomarkers (identifying patients likely to benefit from specific therapies)—provides a key framework for AML biomarker development [34]. Our findings suggest the potential of molecular subtypes as clinically useful tools for precision medicine in drug selection.

4.1 The Independence Paradox: Blueprint vs. Active State

The central innovation of this study is the resolution of the "Independence Paradox": why transcriptomic states can provide highly significant independent predictive value ($p = 10^{-37}$) despite being non-independent for overall survival. We propose that this reflects the fundamental difference between the **Genomic Blueprint** (the mutational potential) and the **Active Lineage State** (the current functional configuration).

While mutations in *TP53* or *NPM1* define the long-term clinical trajectory (prognostic blueprint), the active transcriptomic state—specifically the degree of monocytic differentiation—determines the immediate "engine" of survival and drug vulnerability. This explains why transcriptomics provides critical information beyond genomics for therapeutic selection:

a patient may harbor "favorable" genetics, yet their leukemia blasts have already adopted a resistant, monocytic-like configuration.

Furthermore, our head-to-head benchmarking against the LSC17 stemness signature confirms that lineage state provides independent information to stem cell hierarchy. While the LSC17 score quantifies the "depth" of the stem cell pool, the VRS and its associated clusters capture the "direction" of differentiation. This "Lineage vs. Stemness" distinction represents a second axis of transcriptomic heterogeneity that is essential for a complete understanding of AML biology.

The mechanistic validation through BCL-2 family gene expression provides biological credibility. Higher *BCL2* expression in Cluster 2 is associated with Venetoclax sensitivity through enhanced mitochondrial priming [21, 22]. Conversely, the shift toward *MCL1* and **Oxidative Phosphorylation (OXPHOS)** in Cluster 1 defines a distinct metabolic pathway associated with resistance. This suggests that the monocytic-primed state is not merely a loss of *BCL2* sensitivity, but an active co-option of alternative survival pathways that can be targeted through metabolic or MCL1-specific inhibitors (Supplementary Figure S6).

Upregulation of key immune checkpoints (such as CD47 and CTLA4, which are key components of the VRS) suggests differential immunotherapy responsiveness across subtypes, warranting further investigation of combination approaches [35]. The potential clinical utility of this biomarker is supported by its reliance on routine clinical RNA-sequencing. With the emergence of **rapid RNA-seq protocols** capable of 24- to 48-hour turnarounds in clinical workflows [8], the VRS could potentially be implemented within the critical induction window. Unlike complex multi-parameter panels, the simple tertile-based VRS classification provides an immediate, actionable guide for frontline treatment versus prioritized salvage configurations.

We propose a Phase II clinical trial design (CLUSTER-AML) to prospectively validate cluster-guided therapy. Under this proposed design, the trial would randomize 200 patients 1:1 to cluster-guided treatment (Cluster 1: Panobinostat+Selumetinib+HMA; Cluster 2: Venetoclax+HMA) versus standard of care, with response rate at 3 months as the primary endpoint. This design could enable rapid validation while maintaining equipoise.

4.2 Age-Specific Biology and Pediatric Divergence

The divergent prognostic behavior observed in pediatric AML (TARGET cohort: $HR = 0.81$ ($-p = 0.052$)) highlights the distinct biology of childhood leukemia compared to adult disease.

This finding cautions against inappropriate clinical extrapolation to children and underscores fundamental differences in AML pathogenesis and ontogeny by age [19]. Pediatric AML has distinct mutation landscapes (higher FLT3-ITD, CEBPA, WT1; lower NPM1, TP53) [19], different blast differentiation states, and distinct immune ontogeny. The age-specific effects underscore the importance of age-stratified biomarker development.

4.3 Limitations and Future Directions

This study has limitations. The primary limitation is that drug sensitivity was measured using ex vivo assays on isolated AML blasts, which may not fully recapitulate in vivo response due to tumor microenvironment interactions, drug pharmacokinetics, and patient-specific factors including comorbidities and concomitant medications. While BeatAML ex vivo data have demonstrated correlation with clinical outcomes in prior validation studies [12], prospective clinical validation remains essential before clinical implementation of these findings. Second, the discovery cohort was limited to adult patients; applicability to specific age subgroups (18–40 vs >60 years) requires investigation. Third, we did not assess minimal residual disease (MRD) status, which may modify treatment effects. Fourth, combination therapy predictions (e.g., Venetoclax+Panobinostat) were not tested.

Future work should focus on: (1) prospective clinical trial validation, (2) integration with MRD status and dynamic response monitoring, (3) validation in other AML treatment contexts (intensive chemotherapy, transplant), (4) extension to combination therapy predictions, and (5) development of a CLIA-certified clinical assay.

4.4 Conclusion

Transcriptomic lineage states in adult AML provide predictive value for drug response that is independent of genetic mutations, representing a fundamental shift toward **state-informed, functional precision oncology**. By mapping the "Independence Paradox" and identifying the metabolic and lineage-driven roots of resistance, we provide a promising framework for overcoming therapeutic heterogeneity. This work outlines a translational pathway from discovery to the clinic, helping to identify "hidden" resistance that is not captured by current genomic stratification.

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Figure Legends

Figure 1. Molecular Subtype Discovery and Genomic Landscape in Adult AML. **a**, Principal Component Analysis (PCA) projection of discovery BeatAML cohort ($N = 450$; Cluster 1, $N = 184$; Cluster 2, $N = 266$). **b**, Volcano plot of differentially expressed genes between Cluster 1 and Cluster 2. **c**, Somatic driver mutation frequencies (NPM1, FLT3, TP53, NRAS,

KRAS, WT1, CEBPA) across molecular subtypes. **d**, ELN 2017 risk category distribution across clusters.

Figure 2. The Independence Paradox and Genomic-Orthogonal Predictive Value of Molecular Subtypes. **a**, Multivariable regression variance explained (R^2) showing substantial predictive gains when incorporating transcriptomic subtype over 11 established genomic alterations. **b**, Multivariate regression analysis demonstrating subtype assignment as an independent predictor of Venetoclax response. **c**, Ex vivo Venetoclax response (AUC) distribution in Cluster 1 (Monocytic/Resistant, $N = 89$) vs Cluster 2 (Primitive/Sensitive, $N = 148$; $p = 6.37 \times 10^{-19}$). **d**, Receiver Operating Characteristic (ROC) curve validating predictive specificity for Venetoclax response.

Figure 3. The Monocytic Resistance Axis, Proteomic Target Validation, and Metabolic Reprogramming in Adult AML. **a**, Correlation between bulk monocytic differentiation score and ex vivo Venetoclax sensitivity (AUC) across the BeatAML cohort, establishing the monocytic resistance axis. **b**, Bulk proteomic validation in the matched BeatAML proteomics cohort showing $\log_2$ protein abundance of CD14, MCL-1, and BCL-2 across molecular subtypes. **c**, Malignant blast cell composition fractions (HSC, progenitor, pro-monocyte, monocyte) in Cluster 1 vs Cluster 2 from primary scRNA-seq profiles (van Galen dataset). **d**, Longitudinal single-cell clonal dynamics showing depletion of sensitive and expansion of resistant blast clones from diagnosis to relapse under Venetoclax + Decitabine therapy. **e**, Gene Set Enrichment Analysis (GSEA) of metabolic and hallmark pathways enriched in Cluster 1 (Resistant) vs Cluster 2 (Sensitive).

Figure 4. Single-Cell Resolution and Lineage Dynamics of Venetoclax Resistance. **a**, Single-cell RNA-seq UMAP of primary AML patient samples (van Galen cohort) showing lineage annotations across hematopoietic stem/progenitor cells (HSC/Prog), monocytic, and granulocytic lineages. **b**, Projection of single-cell Venetoclax Response Score (VRS) across single cells showing selective enrichment of high VRS in primitive stem/progenitor cells. **c**, Boxplot of single-cell VRS across cell types. **d**, Expression dotplot of BCL-2 anti-apoptotic family members across single-cell cell types. **e**, Single-cell BCL2 vs MCL1 expression trade-off across lineages.

Figure 5. Development, Multi-Cohort Validation, and Clinical Risk Stratification of the Venetoclax Response Score (VRS). **a**, Distribution of the 9-gene VRS classifier score across discovery BeatAML cohort. **b**, Scatter plot of VRS vs measured Venetoclax AUC

in independent BeatAML 2.0 validation cohort. **c**, Heatmap of ex vivo drug sensitivity profiling across targeted agents. **d**, Stratified clinical overall survival across Low, Medium, and High VRS tiers. **e**, Clinical trial validation of VRS in the VIALE-A cohort.

Figure 6. Actionable Salvage Therapeutics and Rational Combinations for Primary Venetoclax-Resistant AML. **a**, Differential sensitivity of targeted drug classes in Cluster 2 vs Cluster 1. **b**, Mean AUC comparison for top salvage candidates. **c**, Ranking of top 10 differential drugs preferred in Cluster 2. **d**, Dual-combination synergy matrix (SynergyFinder Bliss score). **e**, Panobinostat (HDAC inhibitor) sensitivity boxplot. **f**, Selumetinib (MEK inhibitor) sensitivity boxplot.

Data and Code Availability

The raw and processed sequencing data for the BeatAML cohort are available through the dbGaP repository under accession code phs001657.v1.p1. The TCGA-LAML dataset is available through the Genomic Data Commons (GDC) portal. The pediatric TARGET-AML dataset is available through the NCI TARGET Data Matrix and dbGaP (phs000465). Microarray validation datasets are accessible via NCBI GEO (GSE12417, GSE22845, GSE106291). Single-cell RNA-seq datasets are available under accession GEO GSE116256 (van Galen et al.) and GSE143363.

All custom computational workflows, R/Python scripts, and reproducible pipeline steps are publicly available on GitHub at: https://github.com/rcwang2024/Proj_AML.git. An interactive clinical VRS calculator web application is hosted at: <https://share.streamlit.io/>.

Supplementary Materials

Supplementary Methods, Tables S1–S10, and Figures S1–S16 are available online.

Supplementary Tables:

- Table S1: Sample characteristics across all three cohorts (BeatAML, TCGA-LAML, TARGET-AML)
- Table S2: 50-gene classifier gene list with variable importance and expression patterns

- 789 • Table S3: Master register of study-wide statistical tests and multiple testing corrections
- 790 (40 tests)
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- 798 across independent adult validation cohorts
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Author Contributions

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Competing Interests

The authors declare no competing interests.

Data Availability

BeatAML data are available through dbGaP (accession: phs001657.v1.p1). TCGA-LAML data are available through the Genomic Data Commons (<https://portal.gdc.cancer.gov/>). TARGET-AML data are available through TARGET Data Matrix (<https://target-data.nci.nih.gov/>). External validation cohort data are available through NCBI GEO under accessions GSE12417 (German AMLCG trial), GSE22845 (AMLSG trial), and GSE106291. Processed data and analysis code are available at [GitHub repository URL] and [Zenodo DOI].

Code Availability

All analysis code is available at [GitHub repository URL]. The code repository includes R scripts for consensus clustering, gene signature development, survival analysis, drug response analysis,

825 and figure generation. A Docker container with the complete computational environment is
826 available at [Docker Hub URL].